

Agenda

- Textual tags
- Formatting tags
- List tag
- Table tag
- Resource tags
- Anchor tag
- Form tag
- GET vs POST method
- Header,Footer,section,article,aside,nav Tags

HTML Structure

- It consists of 2 parts

1. Head

- Contains extra information about the page
- tags that can be used inside the head are
 1. title: used to set the title for the tab
 2. script: used to add JS code in the page
 3. style: used to add CSS code
 4. meta: used to add more information about the page
 5. link: used to link external documents (files)
 6. base: used to set the base url used in the page

2. Body

- Contains actual design
- tags that can be used inside the body are
 1. Textual
 2. Resources
 3. List
 4. Table
 5. Linking(Anchor)
 6. Form
- The tags can be inline or block tags
- the inline tags keep the data on the same line however the block tags add the data on the next line

Textual tags

1. Header: used to add header in page
 - There 6 levels
 - Tags: h1 to h6
 - H1 is the biggest while h6 is the smallest
2. Paragraph (

); used to add a para

3. Division()

); used to create groups of Tags and Textual contents

- All the above are block tags

Formatting Tags

- These all are inline tags

1. span - used during CSS
2. bold - `<b></b>` or `<strong></strong>`
3. italic - `<i></i>`
4. underline - `<u></u>`
5. strike - `<strike></strike>`
6. monospace - `<tt></tt>`
7. superscript - `<sup></sup>`
8. subscript - `<sub></sub>`
9. font - `<font></font>`
10. formatted - `<pre></pre>`

HTML Character Entities

	non-breaking space	<code>&nbsp;</code>	<code>&#160;</code>
<	less than	<code>&lt;</code>	<code>&#60;</code>
>	greater than	<code>&gt;</code>	<code>&#62;</code>
&	ampersand	<code>&amp;</code>	<code>&#38;</code>
"	double quotation mark	<code>&quot;</code>	<code>&#34;</code>
'	single quotation mark	<code>&apos;</code>	<code>&#39;</code>
¢	cent	<code>&cent;</code>	<code>&#162;</code>
£	pound	<code>&pound;</code>	<code>&#163;</code>
¥	yen	<code>&yen;</code>	<code>&#165;</code>
€	euro	<code>&euro;</code>	<code>&#8364;</code>
©	copyright	<code>&copy;</code>	<code>&#169;</code>
®	registered trademark	<code>&reg;</code>	<code>&#174;</code>
™	trademark	<code>&trade;</code>	<code>&#8482;</code>

Attribute

- Extra information about the tag
- Attribute always present in name=value format
- E.g. `<meta charset="utf-8">`
 - meta: tag
 - charset: attribute name
 - utf-8: attribute value
- A tag may have one or more attributes
- Every tag has following attributes

- name: used to create query string
- id: used to identify an element uniquely
- style: used to write inline css
- class: used for css

List tags

1. Unordered list

- Does not render the order

```
<ul>
  <li>list item1</li>
  <li>list item2</li>
</ul>
```

2. Ordered list

- Renders the list item order

```
<ol>
  <li>list item1</li>
  <li>list item2</li>
</ol>
```

3. Definition list

- Used to create list of definitions

```
<dl>
  <dt>term 1</dt>
  <dd>definition 1</dd>

  <dt>term 2</dt>
  <dd>definition 2</dd>
</dl>
```

Table tag

- Used to create tabular representation
- Divided into 3 sections

1. Header

- Use `<thead>` (table header)

2. Body

- Use `<tbody>` (table body)

3. Footer

- Use `<tfoot>` (table footer)
- To create row Use `<tr>`
- To create column In header Use `<th>` (table header column)
- In Body and Footer Use `<td>` (table data)
- Example

```
<table>
  <thead>
    <tr>
      <th></th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td></td>
    </tr>
  </tbody>
  <tfoot>
    <tr>
      <td></td>
    </tr>
  </tfoot>
</table>
```

- Attributes:
 1. border: used to create border
 2. colspan: used to merge multiple columns horizontally
 3. rowspan: used to merge multiple columns vertically

Resource tag

1. Images

- img: used to add image
 - src: specify the source
 - alt: used to alternative text (rendered only in case of missing source)

2. Audio

- audio: used to play audio
 - src: to set the file to play
 - controls: to render controls
 - autoplay: to play the file automatically

3. Video

- video: used to play video

- src: to set the file to play
- controls: to render controls
- autoplay: to play the file automatically

Linking Tag (Anchor)

- Used to link multiple pages
 - `<a href="page.html">text</a>`
- multiple sections within same page
 - `<a href="#top">text</a>`
- multiple pages with multiple sections
 - `<a href="page#section">text</a>`

Form tag

- Used to get inputs from user
- Inside the form tag we can use below tags

1. input

- used to get input from the user
- type attribute is used to specify the input type.
- value to type attribute can be
 - text: textual value
 - number: number value
 - email: email value
 - tel: telephone
 - date: date input
 - time: time input
 - radio: create radio button
 - checkbox: multiple selection
 - password: masked characters
 - submit: to submit values
 - reset: to clear the form
 - file: used to upload file

2. textarea

- used to get multi-line input

3. select

- used to render drop-down

```
<select>
  <option>op1</option>
  <option>op2</option>
</select>
```

GET Method

- Values will be sent using Request Head
- Values will be appended on url (visible)
- Restriction on maximum size of data to be passed
- Only ascii values can be sent using GET
- URL can be bookmarked with the values
- Files can NOT sent using GET

POST Method

- Values will be sent using Request Body
- Values will be invisible
- There is no restriction on size of data
- Any value of any type (binary) can be sent by using POST
- URL can be bookmarked without the values
- Files can be using POST

article

- The HTML element represents a self-contained composition in a document, page, application, or site, which is intended to be independently distributable or reusable.
- Examples include a forum post, a magazine or newspaper article, or a blog entry, a product card, a user-submitted comment, an interactive widget or gadget, or any other independent item of content.

aside

- The HTML element represents a portion of a document whose content is only indirectly related to the document's main content.
- Asides are frequently presented as sidebars or call-out boxes.

header

- The element can define a global site header, described as a banner in the accessibility tree.
- It usually includes a logo, company name, search feature, and possibly the global navigation or a slogan.
- It is generally located at the top of the page.

footer

- The HTML element represents a footer for its nearest ancestor sectioning content or sectioning root element.
- It typically contains information about the author of the section, copyright data or links to related documents.

nav

- The HTML element represents a section of a page whose purpose is to provide navigation links, either within the current document or to other documents.
- Common examples of navigation sections are menus, tables of contents, and indexes.
- It's not necessary for all links to be contained in a element.
- is intended only for a major block of navigation links

section

- The HTML element represents a generic standalone section of a document, which doesn't have a more specific semantic element to represent it.

key differences

Tag	Use Case	eg : Newspaper
header	Top part of page/section	Newspaper title
nav	Navigation links	Categories
section	Group of related content	News Category(Sports)
article	Independent content	Individual news story
aside	Side/extra content	Ads/ side notes
footer	Bottom part	publisher info